

Configure Latency Routing Policy in Amazon Route 53

Summary:-

In this lab, a Latency Routing Policy was configured in Amazon Route 53 to route client requests to the AWS Region with the lowest network latency. Two EC2 web servers running in different AWS Regions were associated with latency-based DNS records. The routing policy was then tested to verify that Route 53 automatically directed requests to the endpoint with the lowest latency.

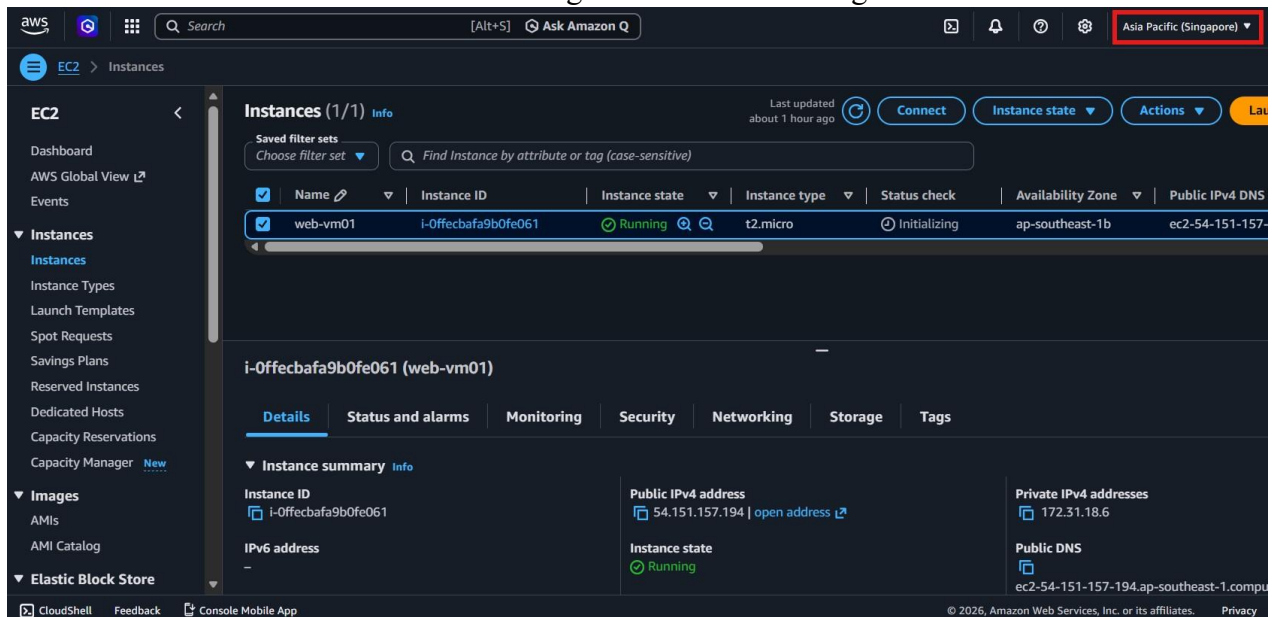
Let's Start:-

- Configure a Latency Routing Policy in Amazon Route 53.
- Create latency-based DNS records for multiple AWS Regions.
- Associate each record with an EC2 web server.
- Verify that Route 53 routes requests to the endpoint with the lowest latency.

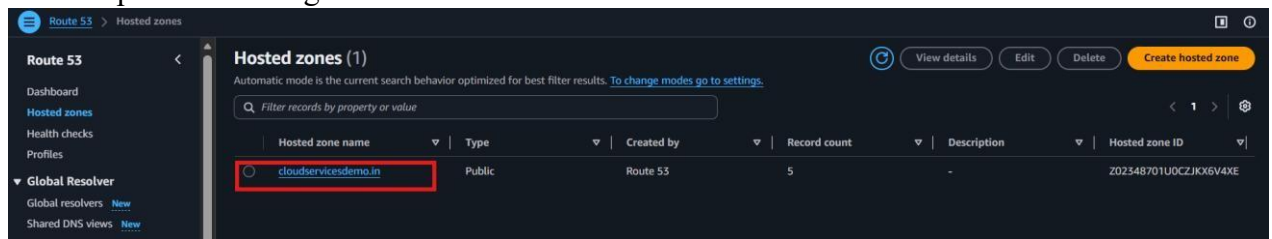
Lab Steps:-

Step 1 – Verify the Environment

- Ensure two EC2 instances are running in different AWS Regions.

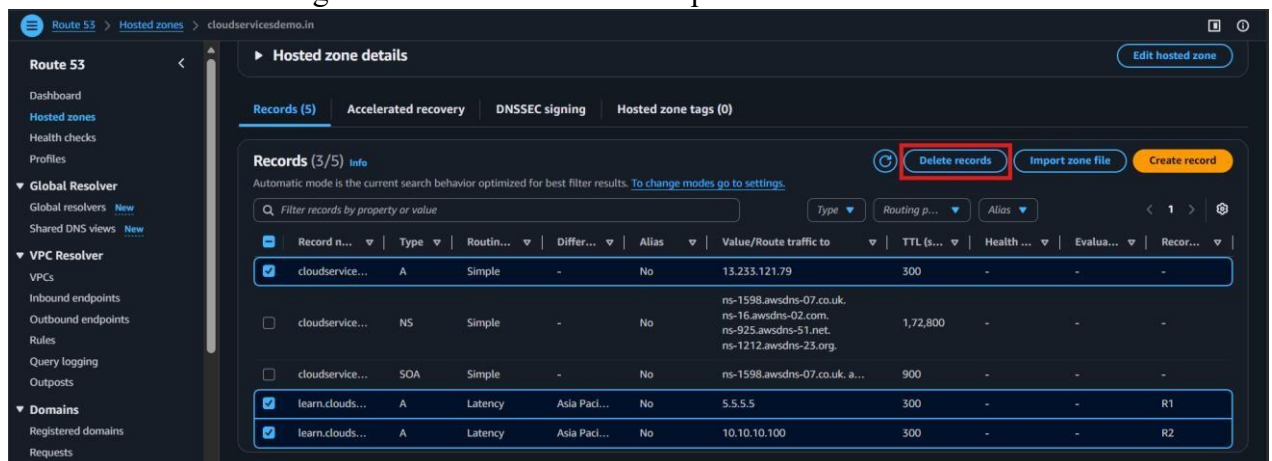


- Repeat the same steps for the second EC2 instance (Singapore).
- Open the existing **Public Hosted Zone** in Amazon Route 53.



Step 2 – Remove Existing DNS Records

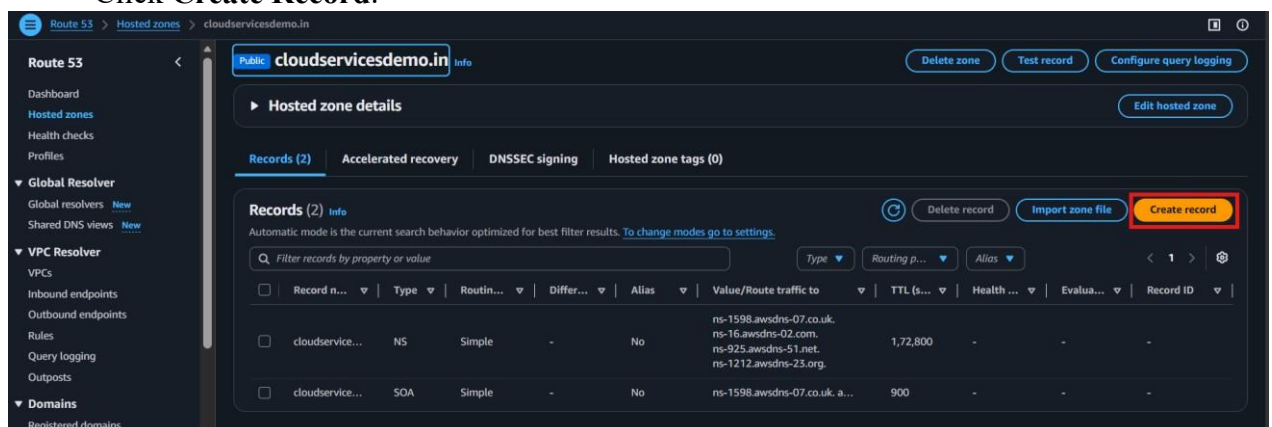
- Select the existing DNS records created in the previous lab.



- Click **Delete Record**.
- Confirm the deletion.

Step 3 – Create the Mumbai Latency Record

- Click **Create Record**.



- Configure the following:
 - **Record Name:** learn
 - **Record Type:** A
 - **Value:** Public IPv4 address of the Mumbai EC2 instance

- **Routing Policy:** Latency
- **Region:** Asia Pacific (Mumbai)
- **Record ID:** Mumbai

The screenshot shows the 'Create record' form in the AWS Route 53 console. The form is for a record named 'learn' in the 'cloudservicesdemo.in' hosted zone. The record type is 'A - Routes traffic to an IPv4 address and some AWS resources'. The value is '13.203.194.240'. The TTL is set to 300 seconds. The routing policy is 'Latency'. The region is 'Asia Pacific (Mumbai)'. The record ID is 'Mumbai'. The 'Add another record' button is at the bottom right.

- Click **Add another record**.

Step 4 – Create the Singapore Latency Record

- Configure the following:
 - **Record Name:** learn
 - **Record Type:** A
 - **Value:** Public IPv4 address of the Singapore EC2 instance
 - **Routing Policy:** Latency
 - **Region:** Asia Pacific (Singapore)
 - **Record ID:** Singapore

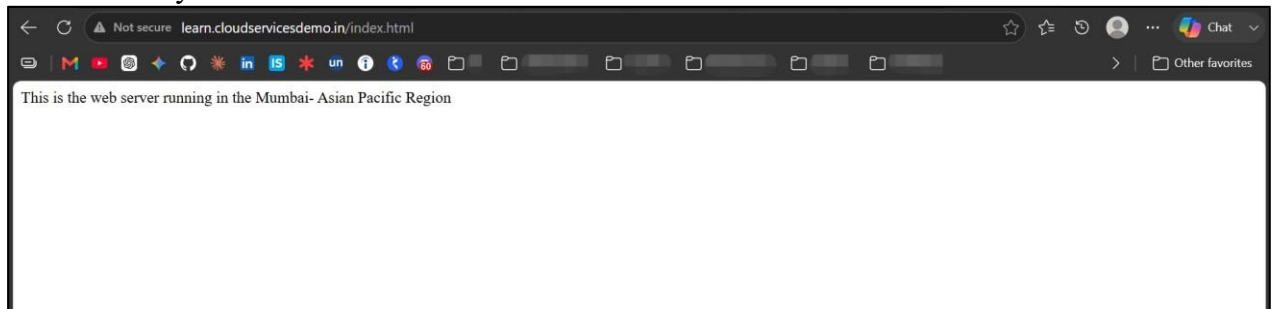
The screenshot shows the 'Create record' form in the AWS Route 53 console for a second record. The record name is 'learn' in the 'cloudservicesdemo.in' hosted zone. The record type is 'A - Routes traffic to an IPv4 address and some AWS resources'. The value is '54.151.157.194'. The TTL is set to 300 seconds. The routing policy is 'Latency'. The region is 'Asia Pacific (Singapore)'. The record ID is 'Singapore'. The 'Delete' button is at the top right.

- Click **Create Records**.

Step 5 – Verify the Latency Routing Policy

- Open a web browser.
- Access:

- learn.<your-domain-name>/index.html
- Verify that Route 53 directs the request to the EC2 instance with the lowest network latency.



Verification

- Configured a Latency Routing Policy successfully.
- Created latency-based DNS records for multiple AWS Regions.
- Verified that Route 53 routed requests to the endpoint with the lowest latency.
- Confirmed successful access to the appropriate EC2 web server.